Vivian Mela

Massachusetts Institute of Technology
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Professors Michel Goemans and Piotr Indyk

Handout 9
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Master Theorem Worksheet

This is a worksheet to help you master solving recurrence relations using the Master Theorem. For each recurrence, either give the asympotic solution using the Master Theorem (state which case), or else state that the Master Theorem doesn't apply. You should be able to go through these **25** recurrences in **10** minutes.

Problem 1-1. $T(n) = 3T(n/2) + n^2$

$a=3$ $\quad$ Crit $= \log_b a$
$b=2$ $\quad$ $= \log_2 3 = 1.58$

CASE 3: $T(n) = \Theta(n^2)$

Problem 1-2. $T(n) = 7T(n/2) + n^2$

$a=7$ $\quad$ Crit $= \log_2 7 \approx 2.8$
$b=2$

CASE 1: $T(n) = \Theta(n^{\log_2 7})$

Problem 1-3. $T(n) = 4T(n/2) + n^2$

$a=4$ $\quad$ Crit $= \log_2 4 = 2 \Rightarrow n^2$
$b=2$

$T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$ $\quad k=0$ $\log^0 n = 1$

CASE 2: $T(n) = \Theta(n^2 \log n)$

Problem 1-4. $T(n) = 3T(n/4) + n \lg n$

$a=3$
$b=4$
Crit $= \log_4 3 = 0.79$ $\quad$ vs $n^1 \log n$

CASE 3: $T(n) = \Theta(n \log n)$

Problem 1-5. $T(n) = 4T(n/2) + \lg n$

$a=4$
$b=2$
Crit $= 2$

CASE 1: $T(n) = \Theta(n^2)$

Problem 1-6. $T(n) = T(n-1) + n$

- recurrence uses subtraction as opposed to division

MT doesn't apply

$T(n) = T(n-1) + n$

$T(n-1) = T(n-2) + (n-1)$ $\quad$ substitute of recurrence

$\rightarrow T(n) = T(n-2) + (n-1) + n$ $\quad$ substitute into org eq.

$T(n-2) = T(n-3) + (n-2)$

$\rightarrow T(n) = T(n-3) + (n-2) + (n-1) + n$

$\rightarrow T(n) = T(1) + 2 + 3 + \ldots + (n-1) + n$

$T(n) = \Theta(1) + \frac{n(n+1)}{2} - 1$

$T(n) = \frac{n^2+n}{2} + \Theta(1) - 1$

... Assume $T(1) = \Theta(1)$

$T(n) = \Theta(1) + \sum_{i=2}^{n} i$

Take the most significant term $\rightarrow$ $T(n) = \Theta(n^2)$

Problem 1-7. $T(n) = 4T(n/2) + n^2 \lg n$

$a=4$
$b=2$
$c_{crit} = \log_2 4 = 2$
n^2

$f(n) = n^2 \log n$
Extended Case 2 $f(n) = \Theta(n^{\log_b a} \log^{k+1} n)$
Master Formula $f(n) = \Theta(n^2 \cdot \log^1 n)$
$k=1$

CASE 3: $T(n) = \Theta(n^2 \log^2 n)$

Problem 1-8. $T(n) = 5T(n/2) + n^2 \lg n$

$a=5$
$b=2$
$c_{crit} = \log_2 5 = 2.32$

CASE 1: $T(n) = \Theta(n^{\log_2 5})$

Problem 1-9. $T(n) = 3T(n/3) + n/\lg n$

$a=3$
$b=3$
$c_{crit} = \log_3 3 = 1$

$f(n) = n/\log n = n \cdot \frac{1}{\log n}$
smaller by dividing by $\log n$ but same polynomial n^1

Case 1 doesn't apply since $f(n)$ isn't polynomially smaller

Problem 1-10. $T(n) = 2T(n/4) + c$

$a=2$
$b=4$
$c_{crit} = \log_4 2 = 0.5$

CASE 1: $T(n) = \Theta(n^{0.5})$

Problem 1-11. $T(n) = T(n/4) + \lg n$

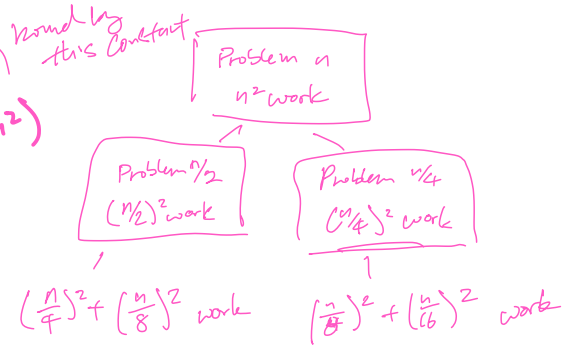
$a=1$
$b=4$
$c_{crit} = \log_4 1 = 0$

$f(n) = \log n$
$T(n) = \Theta(n^0 \log^{k+1} n)$ $k=1$
not exponential

CASE 2: $T(n) = \Theta(n^0 \cdot \log^2 n)$

Problem 1-12. $T(n) = T(n/2) + T(n/4) + n^2$

round log this constant



Doesn't Apply. Recursion tree → $T(n) = \Theta(n^2)$

Problem 1-13. $T(n) = 2T(n/4) + \lg n$

$a=2$
$b=4$
$c_{crit} = \log_4 2 = 0.5$

$f(n) = \log n$
↑ no polynomial

CASE 1: $T(n) = \Theta(n^{0.5})$

Problem 1-14. $T(n) = 3T(n/3) + n \lg n$

$a=3$
$b=3$
$c_{crit} = \log_3 3 = 1$

$f(n) = n^1 \log n$
$T(n) = \Theta(n^1 \log^{1+1} n)$ $k=1$

CASE 2: $T(n) = \Theta(n \log^2 n)$

Problem 1-15. $T(n) = 8T((n - \sqrt{n})/4) + n^2$

doesn't apply bc MT is in the form: $T(n) = aT\left(\frac{n}{b}\right) + f(n)$ $\frac{n-\sqrt{n}}{4} = \frac{n}{4} - \frac{\sqrt{n}}{4}$ ← extra term

MT doesn't apply but $T(n) = \Theta(n^2)$

Problem 1-16. $T(n) = 2T(n/4) + \sqrt{n}$

$a = 2$, $b = 4$, $f(n) = \sqrt{n} = n^{0.5}$, $k = 0$

$\log_4 2 = 0.5$ $T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$ $n^{0.5}$ $\log n$

CASE 2: $T(n) = \Theta(n^{0.5} \log n)$

Problem 1-17. $T(n) = 2T(n/4) + n^{0.51}$

$a = 2$, $b = 4$, $\log_4 2 = 0.5$

CASE 3: $T(n) = \Theta(n^{0.51})$

Problem 1-18. $T(n) = 16T(n/4) + n!$

$a = 16$, $b = 4$, $f(n) = n! = n \times (n-1) \times (n-2)$

$\log_4 16 = 2$

CASE 3: $T(n) = \Theta(n!)$

Problem 1-19. $T(n) = 3T(n/2) + n$

$a = 3$, $b = 2$, $\log_2 3 = 1.58$

CASE 1: $T(n) = \Theta(n^{1.58})$

Problem 1-20. $T(n) = 4T(n/2) + cn$

$a = 4$, $b = 2$, $\log_2 4 = 2$

CASE 1: $T(n) = \Theta(n^2)$

Problem 1-21. $T(n) = 3T(n/3) + n/2$

$a = 3$, $b = 3$, $\log_3 3 = 1$ $f(n) = n/2 = n^1 \cdot \frac{1}{2}$

$T(n) = \Theta(n^1 \cdot \log^0 n)$ $k = 0$

CASE 2: $T(n) = \Theta(n \log n)$

Problem 1-22. $T(n) = 4T(n/2) + n/\lg n$

$a = 4$, $b = 2$, $\log_2 4 = 2$ $f(n) = n^1 \cdot \log n$

$T(n) = \Theta(n^1 \log^1 n)$ $k = 1$

CASE 1: $T(n) = \Theta(n^2)$

Problem 1-23. $T(n) = 7T(n/3) + n^2$

$a = 7$
$b = 3$
$\log_3 7 = 1.77$

CASE 3: $T(n) = \Theta(n^2)$

Problem 1-24. $T(n) = 8T(n/3) + 2^n$

$f(n) = 2^n$ exponential time is greater than polynomial time

$a = 8$
$b = 3$
$\log_8 8 = 1.89$

CASE 3: $T(n) = \Theta(2^n)$

Regularity Condition: $af(\frac{n}{b}) \leq cf(n)$

$8 \cdot 2^{n/3} \leq c\,2^n$

$\frac{8\,2^{n/3}}{2^n} \leq c$

$\frac{a^x}{a^y} = a^{x-y}$

$2^{n/3-n} = 2^{\frac{n}{3}-\frac{3n}{3}}$

$= 2^{-2n/3}$

$8 \cdot 2^{-2n/3} \leq c$

Problem 1-25. $T(n) = 16T(n/4) + n$

$a = 16$
$b = 4$
$\log_4 16 = 2$

CASE 1: $T(n) = \Theta(n^2)$